

DATA ANALYTICS · INTERMEDIATE

Data Analytics Path

Turn raw data into a clear decision.



Turn raw data into trustworthy analysis and decision-ready stories.

🕒 10-16 weeks

📁 4 modules

🎓 Professional learning

Explore the learning experience →

Practical learning. Clear progression. Work you can explain.

YOUR NEXT CAPABILITY

What will you be able to do?

A hands-on path through Excel, SQL, Python, practical statistics, visualization, BI dashboards and stakeholder communication.



Query and prepare data



Build trustworthy dashboards



Apply practical statistics



Communicate decision-ready insights



Who this is for

- Analysts
- Developers moving into data
- Business professionals



Your starting point

- Basic spreadsheet familiarity

Your progression at a glance

FROM CONCEPT TO EVIDENCE



Learn it. Apply it. Explain it.

Move from guided concepts to practical exercises and a coherent piece of work. Use the course resources to revisit difficult ideas and make your reasoning visible.

THE CURRICULUM

A clear route through the subject.

Every module builds towards practical work. The course outline below is aligned with the academy's current program catalog.

1 Analytics Foundations

Questions, metrics and data quality.

- Business questions and KPIs
- Excel analysis
- Data quality
- Analytical thinking

2 SQL and Data Preparation

Query and shape data.

- SQL foundations
- Joins and aggregation
- Cleaning workflows
- Reusable analysis datasets

3 Python and Statistics

Programmatic analysis and evidence.

- Pandas workflows
- Exploratory analysis
- Practical statistics
- Experiment interpretation

4 BI and Storytelling

Dashboards and decisions.

- Visualization principles
- Dashboard design
- Stakeholder narrative
- Analytics capstone

FROM CONCEPT TO EVIDENCE



Practice with purpose

Investigate a practical business question, prepare the data, explore patterns and communicate findings with a clear dashboard and caveats.

BUILD SOMETHING THAT DEMONSTRATES YOUR LEARNING

Decision-ready analytics report

Investigate a practical business question, prepare the data, explore patterns and communicate findings with a clear dashboard and caveats.



Documented queries and cleaning



Interactive dashboard



Evidence-based recommendation



Tools and working methods

Excel

SQL

Python and Pandas

Power BI or Tableau

Tools are part of the learning workflow, not partner endorsements. Examples may evolve while preserving the learning outcomes.



Delivery

Self-paced online · 10–16 weeks

Confirm the current schedule, delivery arrangements and any prerequisites before enrolling.



Completion

A completion certificate is available after the required course work and a passing assessment.

Course completion is not a job, placement or funding guarantee.

Your next steps

- 1 Review the curriculum and your starting point.
- 2 Review current enrollment options and delivery details.
- 3 Start learning, keep your evidence and track your progress.

[Open the course experience →](#)

Customer-preview edition. Interactive links open the local academy prototype. Confirm course availability, schedule and commercial terms with the academy.

INSIDE THE LEARNING EXPERIENCE

Find the denominator before drawing a conclusion

An enrollment report shows 80 completions from one cohort and 60 from another. A stakeholder concludes that the first cohort performed better.

Understand the concept

Counts alone do not establish completion rates. Define the eligible cohort, the observation window and the event used to count completion. Deduplicate learners before aggregation. Keep the definition stable across groups, and separate missing data from a genuine non-completion. Explain uncertainty and confounding factors instead of implying the comparison proves a cause.

Worked example

Illustrative data:

Cohort A: 80 completions / 100 eligible learners = 80%

Cohort B: 60 completions / 60 eligible learners = 100%

A has more completions; B has the higher completion rate.

Check equal follow-up windows before comparing the rates.

Interpret the result

A dashboard should show both numerator and denominator, not only a percentage. The example reverses the initial ranking without changing either completion count. It is teaching data, not a claim about academy performance.

Knowledge check

Why can comparing a new cohort with a fully elapsed cohort be misleading?

Explanation: The new cohort has had less time to complete, so unequal observation windows can bias the comparison.

YOUR PRACTICAL ASSIGNMENT

Build evidence, not just familiarity.

Find the denominator before drawing a conclusion



Your task

- Write a metric definition including grain, time window and exclusions.
- Prepare a deduplicated cohort summary in SQL or a spreadsheet.
- Build a chart with sample size, source date and a decision-oriented annotation.



Submission review criteria

- Correct grain and denominator
- Reconciled totals with documented exclusions
- No causal claim from a descriptive comparison alone

Submit

A working artifact or analysis, a short explanation of your decisions, and evidence that addresses each review criterion.

Reflect

Describe one failed approach, how you diagnosed it, and what you would test next. Identify limitations instead of hiding them.

Your project connection

Investigate a practical business question, prepare the data, explore patterns and communicate findings with a clear dashboard and caveats.

Use the sample assignment as a stepping stone toward the course project, not as a substitute for the full curriculum.

[Explore the worked lesson](#) →

Proposed teaching sample for curriculum and UX review. Full lesson production, instructor review and delivery scheduling remain separate work.